

TrES-1b

R-2248

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_101020 · created 2026-08-15T23:42:14+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455489.66129 +/- 0.00063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.059 +/- 0.036
Transit depth (Rp/Rs)^2	0.35 +/- 0.42 [%]
Orbital Inclination (inc)	86.32 +/- 1.47
Airmass coefficient 1 (a1)	1.204 +/- 0.025
Airmass coefficient 2 (a2)	-0.139 +/- 0.013
Scatter in the residuals of the lightcurve fit is	2.09 %
Best Comparison Star	None
Optimal Aperture	2.34
Optimal Annulus	4.27
Transit Duration (day)	0.069 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.059 $\pm$ 0.036 vs 0.1358 (deviation 2.13 σ)
Duration fit vs published	0.069 d vs 0.1042 d (deviation 1.47 σ)
Mid-time O-C	1.0 min
Depth SNR	1.6
Residual scatter	2.09 %

Light curve

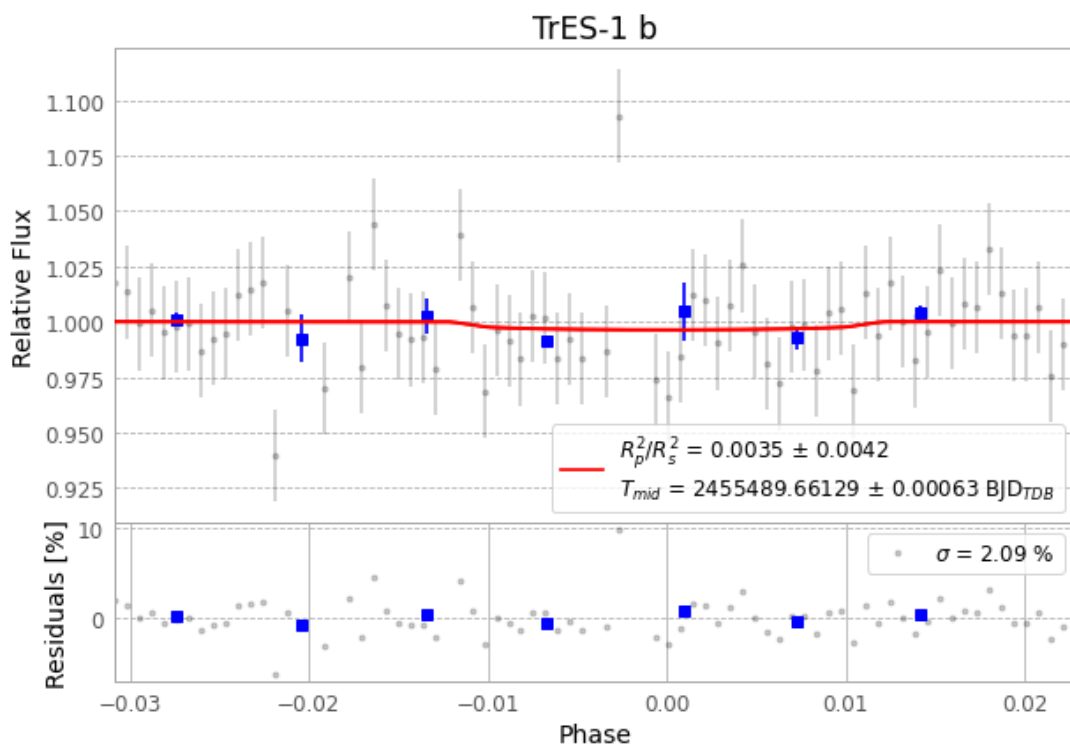


Figure 1 — FinalLightCurve_TrES-1 b_2010-10-19.png (SHA-256 f00bff5e395bb453...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [250, 205], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 06d1f4eee31d2e0d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

79 FITS frames (52 MB), TRES-1101020013614.FITS → TRES-1101020053014.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-101020.log (db0e887d8c0d...), FinalLightCurve_TrES-1 b_2010-10-19.png (f00bff5e395b...), FinalLightCurve_TrES-1 b_2010-10-19.csv (2f063fc028f6...)

Compute resources

Wall time 97.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 23:42Z.